

Methods for Studying Language Development in Children

Group 3

Willemijn Leiner, Emma Linder, Clara Luger, Joel Tauro

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Dr. Rivera

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Introduction

Studying language development processes in children there are many neuroscientific methods available in the field. We are comparing lesion studies, electroencephalography (EEG), transcranial stimulation (TMS), and functional magnetic resonance imaging (fMRI) for studying language development in children. This allows for a better understanding of how the different methods work together to form a holistic understanding. Each method varies in spatial and temporal resolution, invasiveness, cost, and ethical considerations. This shapes the type of questions that can be answered about the development of language processing in children. Communication is vital to humans as species, and those processes form in childhood. We searched databases including PubMed and PsychInfo to find research articles investigating the role of various aspects of the brain during language development in children.

Methods Comparison Analysis

Temporal resolution refers to the precision of a method and how it can measure changes in brain activity over the course of time (Watson & Breedlove, 2019). EEG provides the highest temporal resolution out of all four methods, which allows for recording neural activity in real time and to identify when specific processes related to language occur because EEG records electrical activity directly from the scalp. Researchers can compare electrical patterns with language stimuli to conclude when the brain distinguishes familiar words from just phonemes (Watson & Breedlove, 2019). TMS has good temporal resolution, but it isn't as fast as EEG, which is able to differentiate electrical activity by milliseconds. A magnetic pulse is given to the scalp and can be timed to occur when a person sees, hears, or responds to something. TMS can give a good idea into how quickly a behavior is disrupted, but not when functions on a neuronal level are disrupted (Watson & Breedlove, 2019). FMRI has poor temporal resolution, since the

blood-oxygen-dependent (BOLD) signal is measured, which doesn't directly impact the neural activity. The hemodynamic response lags seconds behind the actual firing of the neurons, which makes it ineffective at tracking rapid processes. It is still valuable for examining longer sequential processes like comprehension (Watson & Breedlove, 2019). Lesion studies do not have any temporal resolution because they measure impaired processing long after damage has occurred.

Spatial resolution, on the other hand, describes how accurately a method is able to locate where in the brain the activity occurs (Watson & Breedlove, 2019). fMRI is the most accurate method in this category, because it provides detailed images of the blood oxygen activation at an extremely small scale, which helps the researchers precisely localize brain activity in regions such as the Broca's and Wernicke areas. Lesion studies also have relatively high spatial specificity. By examining which areas are damaged, and how the damage affects comprehension or speech, researchers are able to infer which regions are used for language. However, results can be varied widely since each lesion is unique, making it difficult to generalize the findings. TMS has moderate spatial resolution. It can only focus on small patches of the brain near the surface (usually about one to two centimeters), and they are not able to reach deeper brain structures. TMS can also temporarily disrupt activity in a region, which can be useful in the process of testing whether a specific area is important in a function like language processing. EEG has very poor spatial resolution. The electrical signals that travel through the skull and scalp are distorted which makes it extremely difficult to identify where exactly it is coming from. When the number of electrodes is increased, it can be slightly more precise, but EEG is mainly suited for temporal resolution rather than location. EEG data is also prone to artifacts due to the many layers that the electrical signal goes through. Electrical signals from the brain must travel through the cerebral

cortex, skull, skin, and hair before they reach the electrodes used to measure the signal. By the time it reaches the electrodes, signals may be distorted or even decayed completely.

All four methods are non-invasive, though they do vary in levels of discomfort or risk. EEG is both non-invasive and painless; it only involves small electrodes placed onto the scalp to measure electrical brain activity from the surface. fMRI is also non-invasive because it uses a magnetic scanner to create a functioning image of the brain. TMS temporarily disrupts the brain by directly stimulating it with magnetic pulses, but is still considered non-invasive. Lesion studies are minimally invasive, they rely on preexisting injuries rather than experiments. Collection of data usually includes neuroimaging and behavioral testing.

fMRI is the most expensive, since scanners cost millions of dollars and require specialized facilities leading to limited accessibility for smaller/resource limited institutions. Each scanning session costs multiple hundreds of dollars. TMS, like fMRI, has costly equipment and requires specialized researchers & facilities to run. EEG systems cost a fair amount of money to purchase initially, but running sessions can be relatively inexpensive compared to methods like fMRI. The main cost of lesion studies comes from a costly structural brain scan like an MRI, but otherwise running cognitive & behavioral tests is relatively affordable.

Ethical considerations are especially important when studying young children's language development. EEG and fMRI require attention to the participants' comfort, but will pose no risk. For fMRI data collection, fears of confined spaces and loud noises need to be addressed. Due to potential side effects, TMS involves stricter oversight to meet the safety guidelines and informed consent requirements. TMS can trigger seizures, so proper training must be ensured before researchers conduct TMS on participants. Lesion studies cannot directly control the type of lesion a patient has on ethical grounds. Researchers can only study willing participants with

pre-existing damage. Due to the lack of control, lesion studies cannot ethically be used to infer causation. Once informed consent is obtained for a lesion study, however, ethical risks are typically minimal. There are no extra ethical requirements for EEG studies outside of the general requirements of informed consent, do no harm, and confidentiality of participants. There is no risk of harm associated with EEG studies.

Synthesis

Lesion studies can often be a first step in identifying crucial brain regions for a certain process, such as language development, to occur. Na et al. (2022) used VLSM mapping in a meta-analysis to identify brain regions that most often lead to language impairment, like the left posterior superior temporal regions (Wernicke's area) for language comprehension, and the inferior frontal region (Broca's area) for speech production. Sun et al. (2024) shows how lesion studies can provide a more complete picture because it shows how dysarthria can also disrupt language functioning. Other methods may neglect the importance of neuromuscular functioning because they are hyper-focused on brain structures. Strengthening these theories, Bonhage et al. (2015) studied the brain areas activated by sentence comprehension, and found that similar areas were activated during language processing. Another fMRI study, Plante et al. (2006) found right hemisphere activation when children processed speech intonation and rhythm. TMS can then provide something lesion studies and fMRI studies largely cannot: *causal* evidence. TMS allows researchers to temporarily impair areas and measure subsequent disruptions. Rejnö-Habte Selassie et al. (2018)'s TMS experiments were able to provide causal evidence that a significant portion of language processing & development in children occurs in the left perisylvian region, consistent with the fMRI and lesion study findings.

Another crucial component to understanding how language works in the brain is temporal analysis of when specific brain activity occurs. A study by Kwok et al. (2018) used resting-state EEG recordings to measure alpha oscillations and oral language ability in children between the ages of 4 and 6. EEG can demonstrate how alpha oscillations and language processing speed both increase with age. Furthermore, an fMRI study was used to determine what areas of the brain are used to predict aspects of language. This suggests that within the language prediction temporal window, there are specific language systems used to anticipate future words based on grammar and vocabulary patterns (Shain et al. 2020). A study combining TMS and EEG investigated when specifically these predictions occur within the sentence structure. The results found that TMS had a short-lasting effect as a result of the mid-sentence verb (Krocze et al. 2019). This suggests that temporal areas play a later role than frontal brain areas in predicting the upcoming noun.

In addition, combining various brain imaging tools can help researchers gain a more holistic understanding of factors that may disrupt normal language development. This includes determining when, where, and why brain development and language activity deviate from the “norm.” A lesion study by Coleman et al. (2016) of preschool-aged children with cerebral palsy found that cortical gray matter lesions impaired communicative abilities more than periventricular white matter lesions, suggesting that disrupting neuronal bodies is more detrimental than the myelinated connections, and that gray matter growth is more important to language development in children. An EEG study by Maguire & Schneider (2019) showed how socioeconomic status affected children’s alpha power and theta levels during their resting state. This corresponded with a lower language vocabulary and working memory in children from low

socioeconomic backgrounds compared to children of the same age and gender from high socioeconomic backgrounds.

These methods can be used to promote language development as well. Research has shown that children with autism spectrum disorder developed greater progress in expressive language, responsiveness, and measures of social engagement as a result of active repetitive transcranial magnetic stimulation to Broca's area as compared to the control group (Darwish et al. 2021). Another study found that a combination of transcranial direct current stimulation and language-cognitive training were effective in improving language and cognitive ability in children aged 2-6 with language development difficulties (Zhou et al. 2024). Studying the brain can not only identify where developmental complications occur, but also can be used to help mitigate these deficits.

EEG studies have shown that infants' cognitive abilities and development are highly correlated with age (Bradley et al., 2024). However, there are some situations that may cause a child's development to deviate from typical patterns. A benefit to studying language development in children is that younger brains can display more neuroplasticity if a process is disrupted at an early age. Malaia et al. (2020) found that participants' age of sign language acquisition was significantly negatively correlated with the mean N400 amplitude response when measuring sign language capabilities. This shows that language development is still fluid in young children, and Sreedharan et al. (2018), an fMRI study, strengthens that hypothesis. In visual verb generation and word pair tasks, most participants displayed dominant left-hemisphere activation, but some displayed bilateral activation, showing that language lateralization continues to develop later in childhood. Lesion studies are especially useful to analyze neural plasticity because they can study long-term neural adaptation that other methods cannot. Chilosi et al.

(2005) found that despite children with left hemisphere damage (LHD) consistently scoring worse on language tasks than children with RHD, they displayed atypical lateralization in activation during language processing, so they did not lose all of their language capabilities.

Another way one can study neural plasticity is through studying potential treatments. Children and adolescents diagnosed with developmental dyslexia who received active tDCS showed superior improvements in reading speed, accuracy, and phonological processing compared to those who received sham stimulation (Costanzo et al., 2016). This can be seen as a next step in researching language development in children, because now methods such as TMS can causally measure effective treatments caused by lesions or impairments that EEG or fMRI pick up on. All of these methods are best used in conjunction with one another, as a meta-analysis by Prentice et al. (2025) demonstrates. They used fMRI to study BOLD activation in brains with epileptic lesions to attempt to discover a concrete age in which atypical lateralization cannot occur. This study shows how using multiple methods for one research question can provide better insight into a research question. It is also important to note that without methods like fMRI, it is more difficult to measure neural plasticity in lesion studies, showing how triangulation of methods occurs even in studies focused on one technique.

There are valuable insights but also constraints within each method. Lesion studies aren't able to ethically establish causation in humans and they often rely on naturally occurring accidents or injuries. TMS and tDCS's are limited to cortical regions, and they are not able to reach subcortical structures. EEG measures only electrical activity on the surface level with low spatial resolution. While fMRI has great spatial resolution, it has limited temporal resolution, and may thus miss rapid timing of language processing. No single technique is sufficient working alone, and integration across the modalities provides a more complete picture.

Future research should focus on defining more precise critical periods for certain language functions in children because current studies only provide broad developmental windows. Subcortical contributions to language also remain quite unexplored, due to cortical limitations of EEG and TMS. The combination of an effective spatial method like fMRI, a method with exemplary temporal resolution like EEG, a causal method like TMS, and a long-term lesion study will provide a more full understanding of how language emerges throughout the development.

Conclusion

The integration of lesion studies, EEG, fMRI, and TMS provides a comprehensive understanding of how language develops in the brain. When all of the techniques are used together, the approaches reveal the structure and timing of language development, but also the brain's capacity for adaptation and recovery. The continuation of collaboration between methods will allow researchers to better understand critical periods, subcortical contributions, and apply this knowledge to improve language outcomes in children.

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